

Concept-based behavioral planning and the lateral prefrontal cortex

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Many lines of evidence implicate the lateral prefrontal cortex (LPFC) in the executive control of behavior. In early studies, neuronal activity in this area was thought to retain information about forthcoming movements for a short period until they were executed. However, later studies have stressed its role in the cognitive aspects of behavioral planning, such as behavioral significance, behavioral rules and behavioral goals. The consequence of the intended action (i.e. a change in the state of the target object), rather than the intended movement, is primarily represented in the LPFC during planning. Recent studies show that the LPFC is involved in more abstract aspects of conceptual processes, such as in representing categories of multiple actions at the stage of behavioral planning.

Introduction

The lateral prefrontal cortex (LPFC) possesses a wealth of anatomical connectivity with multiple cortical and subcortical areas [1,2] and is involved in broad aspects of behavioral control [3–5]. Classical theories on the function of the LPFC have centered on working memory [6,7], attentional control [8] and behavioral inhibition [9]. Recent studies of this area, however, have revealed its role in the control of a much broader spectrum of behaviors [10–12]. Beyond the short-term retention of information, much interest has grown in its role in cross-modal and cross-temporal association of information [13], in the executive control of behavior [14,15], and in the top-down control of neural networks involving the cortical and subcortical areas [16]. This article, based on recent physiological findings, focuses on the involvement of the LPFC in behavioral planning that requires information processing at the conceptual level, and is not intended to provide a comprehensive review of studies on the LPFC.

Behavioral planning in the prefrontal cortex

Because of its location in the frontal lobe, and its efferent connections to higher-order motor areas, the LPFC plays important roles in regulating aspects of volitional motor behavior [3,5,17]. Therefore, it is rather surprising that relatively few experimental studies have explored its role in the motor aspects of behavior compared with the number of studies on perceptual, mnemonic and attentional aspects of behavior. Studies using functional magnetic resonance imaging (fMRI) in humans have demonstrated

prefrontal activity that represents forthcoming actions [18–20]. Studies examining single cells of the LPFC in macaques have also reported activity that might represent the parameters of forthcoming movements [21]. This raises the question as to whether intended movements are represented in the LPFC in ways that are similar to how they are represented in the cortical motor areas.

A series of experiments has revealed that behavioral factors other than the specification of variables for intended movements crucially influence neuronal activity in the LPFC. Hoshi *et al.* [22] found that neuronal activity (recorded from the periprincipal portion of the LPFC) during a preparatory period differed considerably depending on whether the monkeys were intending to capture a circle or a triangle appearing in a specific spatial location. Furthermore, they found that activity in the LPFC during the preparatory period differed greatly depending on the behavioral rule that specified the intended motor task. Neuronal activity during the preparatory period might reflect a shape-match or location-match rule [23], a matching-to-a-sample rule or a spatial-match rule [24], or a match or non-match to sample rule [25]. Human studies in this direction appear in a comprehensive review [26].

In a more recent study, the issue of the nature of motor representation within the LPFC was addressed by designing a behavioral condition that dissociated two behavioral factors specifying either motor variables (i.e. arm movement) or the motion of an object to be acted upon [27]. Monkeys were trained to move a cursor on a video screen by two hand-operated manipulanda. Supination (palm up) or pronation (palm down) of the right or left hand moved the manipulanda in one of four directions, which were then linked to the movements of the cursor (up, down, right or left). The direction of the cursor's motion was assigned to each of four different hand movements in three different ways. It was found that, during the preparatory period, the vast majority of LPFC neurons exhibited activity reflecting the movement of the cursor, but not the movement to be performed (i.e. use of the right or left hand, or the direction of the movement). In other words, the LPFC neurons primarily represent the movement of an object that will occur as a consequence of an intended limb movement, but not the limb movement *per se*. This raises the possibility that motor behavior planning in the LPFC is performed in terms of the end result, but not in terms of the intended movement.

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Averbeck *et al.* [28] studied the LPFC while monkeys were required to copy geometrical shapes from the right half of a screen to the left half of the screen. They reported that the ensemble activity of LPFC neurons reflected sequential segments of movement being planned. However, this interesting report could also be interpreted as if the neuronal activity reflected the shape of the geometric figure that the monkey intended to draw, rather than reflecting the limb movements planned. In this experiment, these two behavioral variables were not dissociated. It may well be that the behavioral planning in the LPFC is achieved in terms of the end result (i.e. an object intended to be drawn), rather than in terms of the movements required to perform the task.

Identifying behavioral goal independent of reward contingency

The LPFC is influenced by information concerning the quality, quantity and content of reinforcement that is expected as the outcome of a specific behavior [29], although the orbitofrontal and medial prefrontal cortices are the principal divisions involved in reinforcement-guided decision making [30]. The primary importance of reinforcer-related activity in the LPFC might well be the modulation of behavioral control, reflecting the level of attention and motivation contingent on the expected positive-negative reinforcer [31]. The term behavioral goal, or goal-based action selection, is sometimes used to signify the presence or absence of a reward as the outcome of an action [32]. However, in view of the ubiquitous nature of reward expectancy in influencing neural activity throughout the central nervous system, we propose a more limited use of the term 'behavioral goal' when discussing prefrontal functions, referring specifically to gaining access to an object or place or attaining a specific behavioral state in a given environment, independent of reward relevance.

The neglect of behavioral goals is a hallmark of frontal lobe deficiencies [33,34]. Frontal patients often fail to achieve the behavioral goal during tasks that require cognitive control such as multistep tasks or the concurrent execution of dual tasks [35,36]. When attempting the Tower of Hanoi task [37,38], the prefrontal cortex is implicated in resolving goal-subgoal conflicts. Such clinical studies of and reports on human subjects prompted researchers to determine the representation of behavioral goal information in the PFC in experimental studies. Saito *et al.* [27] constructed a spatial maze requiring monkeys to achieve multiple behavioral goals in a temporal sequence (Box 1). During a preparatory period, the monkeys were required to plan a path to acquire intermediate goals stepwise in order to reach the final goal. During the planning period, neuronal activity in the LPFC initially represented the final goal and subsequently represented the first, and then the second, of the intermediate goals. Interestingly, the information specifying these intermediate-goal representations was not available in externally given instruction signals, but instead was generated internally. These findings demonstrate that LPFC neurons take part in governing goal-oriented behavior by the generation of goal and subgoal representations while conforming to an appropriate behavioral schedule (see Box 1).

Box 1. Representation of behavioral goals explored during a path-planning task

In a study designed to explore the neural representation of behavioral goals in the LPFC [27], monkeys were trained to perform a path-planning task in which multiple movements of a cursor were required to reach a goal within a maze (Figure 1a). A checkerboard-like maze and a cursor were displayed on a screen. The movement of the cursor was linked to the movement of manipulanda. At the beginning of the task, the animal was required to hold the two manipulanda in a neutral position for 1 s until a cursor appeared at the center of the maze, indicating the starting position. One second later, a goal position was presented for 1 s (goal display). After a delay period, the color of the start-position cursor changed from green to yellow, which served as a task-initiation signal (first GO signal). The animal was required to move the cursor within 1 s, using the manipulanda. The cursor moved in the direction specified by the animal to a position that was defined as the immediate goal. After holding the cursor there for 1 s, the next GO signal was presented. The animal was required to move the cursor in a stepwise manner to reach the goal that had been presented during goal display. No visual cues were provided to indicate the direction in which the cursor should be moved during the task. Thus, the monkey selected the direction of each cursor motion based solely upon its intended moves to reach the final goal. When the cursor reached the final goal, the animal received a reward. The population analysis of prefrontal neurons revealed that neuronal activity reflecting the final goal rose sharply at first and then subsided to a sustained level (Figure 1b, red line). In the later part of the delay period, during which the monkey was still formulating a plan, activity reflecting the immediate goal gradually increased and then peaked before the initiation of the first move (blue line). This finding implicates the prefrontal cortex in planning goal-oriented sequential behavior.

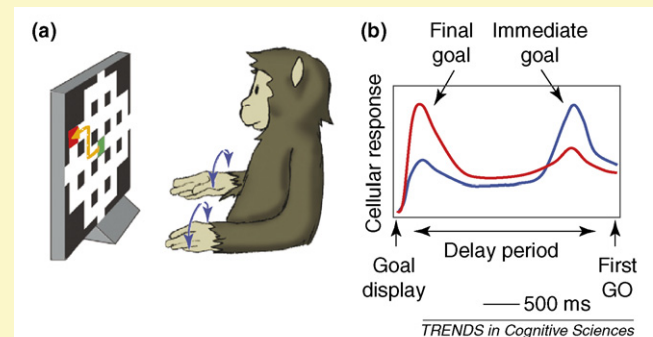


Figure 1. (a) Experimental set up. Each monkey faced a maze on a screen and was required to formulate a stepwise plan to move the cursor from the starting position to the final goal using two hand-operated manipulanda. The direction of cursor movement on the screen was linked to the movements of the manipulanda in a variable manner. **(b)** Plots of cellular responses representing the positions of the final goal (red) and the immediate goal (blue) on the screen. Neural activity during the delay period (behavioral planning period) was calculated from the population response of prefrontal cells and plotted as the intensity of information-content values on the ordinate. (Adapted, with permission, from Ref. [27].).

Behavioral sequencing in the frontal cortex

Much of purposeful behavior depends on the execution of multiple actions arranged in an appropriate temporal order [39]. Cortical and subcortical mechanisms regulating temporal organization of actions have been explored extensively [40]. Barone and Joseph [41] found that neuronal activity in the LPFC depended on the sequence of a visuospatial eye-hand task during which monkeys were reaching for three spatial targets located at the top, right and left of a panel. Neuronal activity during the instructed

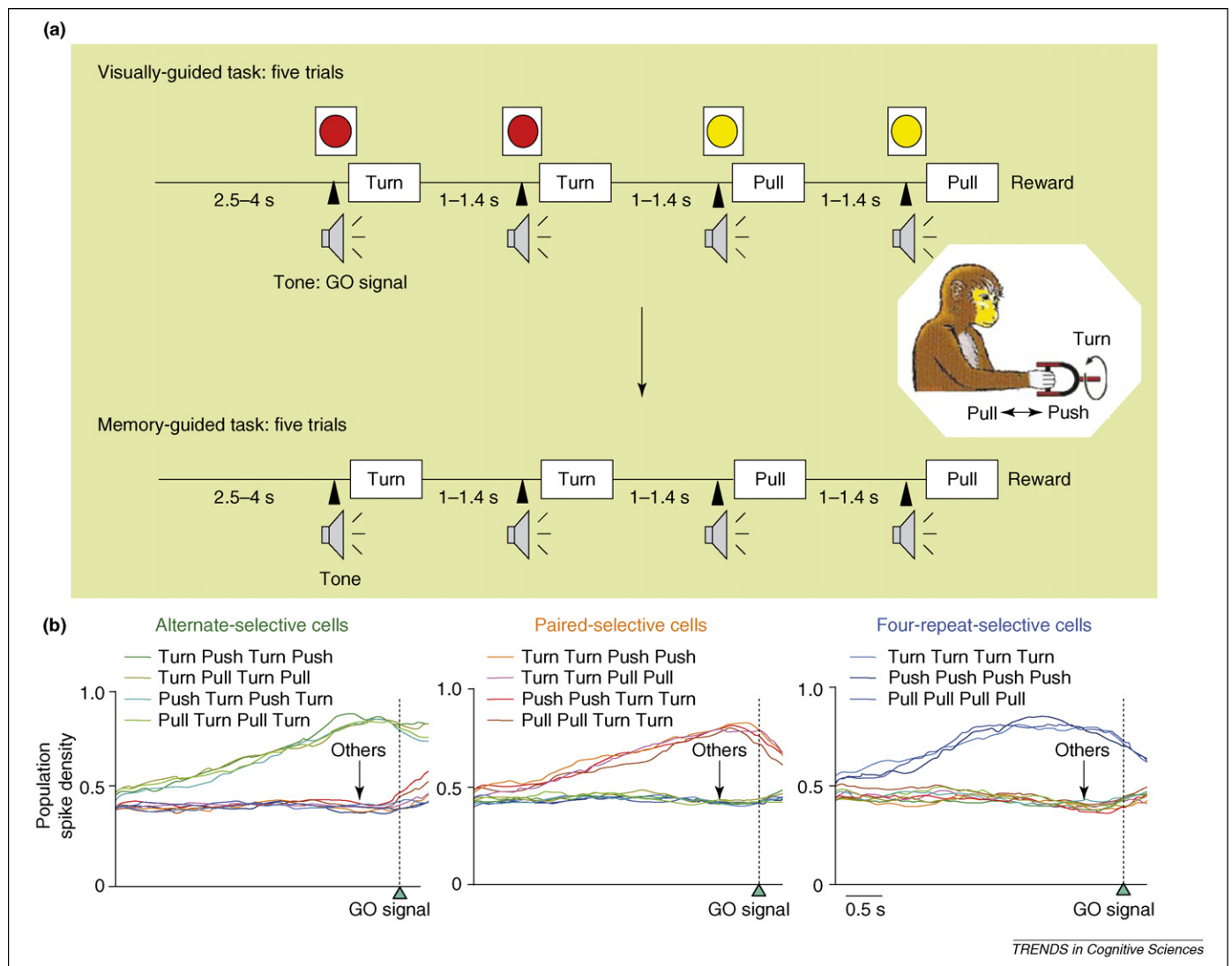


Figure 1. Behavioral categorization task and category-selective neuronal activity. **(a)** Diagram illustrating the temporal sequence of events in a behavioral task used to examine the role of the prefrontal cortex in behavioral programming requiring the categorization of multiple sequences. Initially, a correct sequence of four movements was presented using visual signals. Monkeys were allowed five trials under visual guidance and were then required to perform the same sequence based on memory. After the completion of five trials by memory, the sequence of four movements was renewed in subsequent visually guided trials. **(b)** Time courses of prefrontal cell population response showing category selectivity. The spike density function for prefrontal cell populations shows those that were selectively active while the monkey prepared behavioral sequences from one of the three categories. The traces compare activity during preparation for alternate (left), paired (middle) and four-repeat (right) sequences with activity recorded during preparation for the sequences of other categories (adapted, with permission, from Ref. [45]).

delay period and during the execution period was specific to the sequence of the targets to be captured. Because this task required eye–hand movements with specific trajectories in space, neuronal activity might have signaled selectivity for the spatial trajectory of movements, rather than for the temporal sequence itself. In studies that minimized spatial trajectory as a confounding factor, medial premotor areas (i.e. the supplementary motor area [SMA] and pre-supplementary motor area [pre-SMA]) are found to be profoundly involved in the temporal organization of multiple actions [42,43]. Neuronal activity constituting building blocks of time structures that enable sequential actions have been found in medial motor areas of monkeys [40,42–44]. Such neuronal activity specifically coding temporal-sequence information was not prominent in the LPFC in a recent report [45] (see below). Differences in the task requirements seem to be decisive in shaping how LPFC neurons are recruited for sequential coding. If

spatial-sequence factors are crucial, then the participation of the LPFC is likely to be more apparent [46,47]. A recent fMRI study [48] found that the human LPFC was particularly active when subjects were engaged in selecting task-relevant elements from chunks of numbers (segmenting operation), rather than when engaged in converting previously unrelated elements into a sequence (binding operation).

Behavioral categorization and concept-guided behavioral planning

In this section, we deal with behavioral planning based on the development of a generalized concept for a set of behavioral activities encompassing multiple events. When confronted with the need to plan multiple actions in several different manners, behavioral planning begins with the formulation of a general plan, followed by the individual details. To formulate a general plan, the categorization of a

series of actions is central to achieving the objective, and clinical studies have implicated the prefrontal cortex in such processes. An aspect of the behavioral problems observed in patients with frontal cortex lesions is the failure to achieve the objective of goal-oriented behavior through a series of simple actions [49]. Frontal patients also have difficulty arranging a set of simple, open-ended tasks in an appropriate temporal order to achieve a behavioral goal [50]. More recently, clinical reports have suggested that patients with prefrontal lesions show impairment when analyzing clusters of action sequences [51] or when formulating a coherent and structured action plan [52,53].

To investigate the involvement of the LPFC in formulating macro-level plans that regulate the sequential order of multiple actions, an experimental model was recently introduced with which to analyze neuronal activity in the LPFC during the conceptual formulation of a plan, including the categorization of planned behavior [45]. When subjects are required to remember a large number of complex motor sequences and execute each of them individually, planning is much easier if multiple sequences are categorized according to the specific temporal structure included in each subset of sequences. In the experimental model incorporating this basic assumption, monkeys were trained to remember 11 different sequences for performing four actions (Figure 1a). The individual sequences belonged to one of three categories: paired sequences containing two pairs of actions (e.g. push–push–turn–turn); alternate sequences with two alternating actions (e.g. turn–pull–turn–pull); and four-repeat sequences (e.g. push–push–push–push). Behavioral analysis revealed that the monkeys depended on this categorization to remember each of the 11 sequences. Cells in the LPFC

exhibited activity representative of a specific category of behavioral sequences. Categories of behaviors, embodied by different types of movement sequences, were represented in prefrontal cells during the process of planning (Figure 1b). As many as 51% of the cells that were active during the planning period exhibited such selectivity for the category. Interestingly, neuronal activities that were specific to individual action sequences were in the minority. These findings indicate that macaque monkeys use categorized information concerning temporal patterns of motor behavior (cf. [54]) and that LPFC cells specify behavioral categories during planning. This cellular activity represents the macro-level structure of planned sequences, exemplifying the development of behavioral planning at a conceptual level in the LPFC. This view is consistent with the hypothesis that temporal structures that characterize common patterns among event complexes are represented in the LPFC [55].

Categorization is at the core of concept formation (Box 2), and the categorization of visual stimuli is represented in the LPFC [56]. LPFC neural activity was examined in monkeys that had been trained to categorize animal forms as being either ‘dog-like’ or ‘cat-like’. When relating the basic form of one animal to that of the other animal, LPFC cells selectively responded to visual stimuli belonging to either the cat or the dog category. Shima *et al.* [45] found an entirely new type of category representation in the LPFC with respect to the temporal structure of multiple motor behaviors during behavioral planning. As previously mentioned, they identified neuronal representation for three different action pattern categories: alternate, paired or repeat. We propose a conceptual flow of processes (Figure 2) in which neural structures use category information to plan multiple sequences of actions.

Box 2. Categorization and the lateral prefrontal cortex

Living systems must categorize; objects and events in the environment are assigned to separate classes or categories [61,62]. Categorization allows the recognition of old or new entities without the need for new associative learning. Numerous brain structures are involved in different types of category learning [63,64]. Although brain imaging studies in humans have determined which brain areas are involved in different classes or categories [65–68], studies in non-human primates have revealed that individual cells take part in representing category information. In the lateral prefrontal cortex (LPFC), cells responded selectively to a class of objects based on shape resemblance [56]. In this study, prefrontal cells not only differentiated between ‘dog-like’ or ‘cat-like’ shapes but also responded differently to two classes of objects whose shape boundaries were altered by the experimenters. Thus, category selectivity was based on the similarity of shapes, and category learning was based on explicit training with reinforcement. Furthermore, these category-selective responses were coupled to two different actions: match response (GO) and non-match response (NO-GO). By contrast, category selectivity could also be based on the temporal structure of actions to be performed [45]. In this case, subjects learned categories through a procedural learning process; no external cues were given to prompt the development of category-selective responses. The appearance of category selectivity was not immediately coupled to any reward-laden motor responses. These studies demonstrate that different classes of categories learned in different behavioral contexts are represented in the LPFC to guide discriminative behavior and to plan complex behavioral sequences.

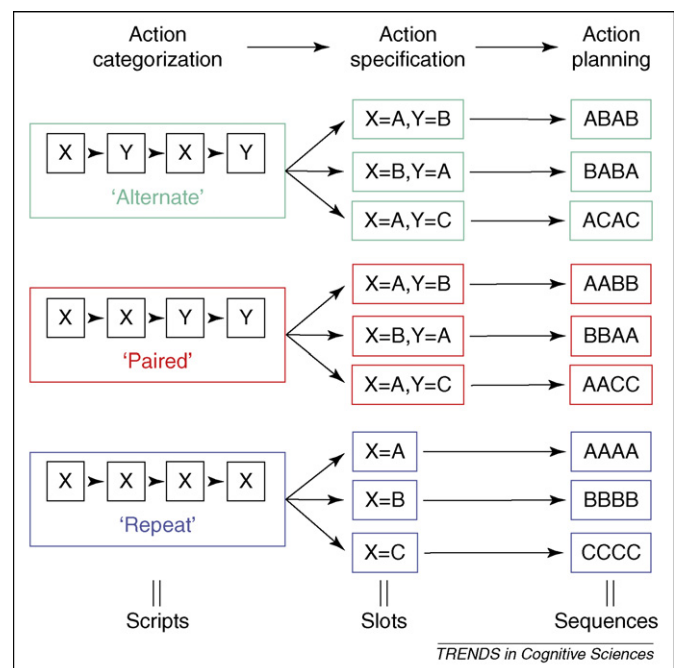


Figure 2. Conceptual flow of processes in which neural structures use category information to plan multiple action sequences. Action categories serve as ‘scripts’ that describe a general structure for an appropriate sequence of events. In the next stage of planning, specific items in a script are identified (i.e. if each slot in a script is identified), and information for an appropriate action sequence is available.

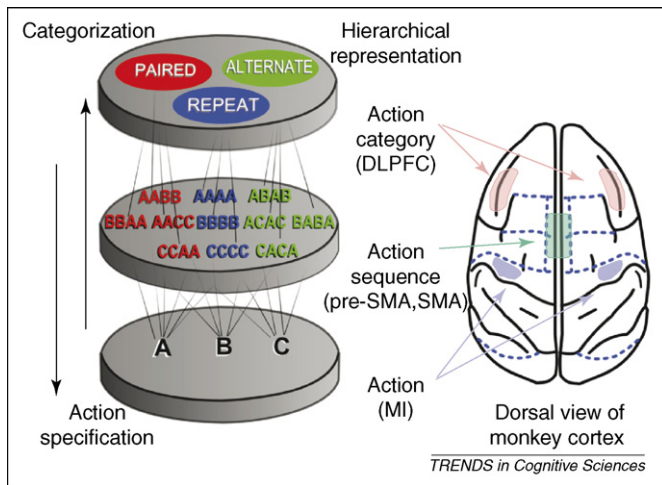


Figure 3. Schematic diagram demonstrating the hierarchical organization of neural structures specifying individual actions (bottom), sequences of actions arranged in different temporal orders (middle), and categories of sequences encompassing subsets of sequences according to common structural patterns (top). Action categories are primarily represented in the dorsolateral prefrontal cortex (DLPFC), whereas action sequences are represented more in the supplementary motor area (SMA) and pre-SMA. Individual actions are primarily specified in the primary motor cortex (MI). Approximate locations of the DLPFC, medial premotor areas, and MI are indicated in the monkey cerebral hemispheres (right inset).

Action categories serve as ‘scripts’ [57,58] that describe a general structure of appropriate event sequences. When, in the next stage of planning, specific items in a script are identified (i.e. if each slot in a script is identified), information for the appropriate sequence of actions is available. Such representation of higher-order structure might have emerged out of the interaction between simpler processing elements, developing gradually through learning, in a manner that might require recurrent connections within neural networks [59]. Through the development of such interactive networks, monkeys might have learned the abstract structure as well as the surface structure of sequences during extensive training [60]. Although the LPFC neurons representing each category were frequently identified, it is interesting that neuronal activity representing individual action sequences, which is abundant in medial premotor areas, was not common in the LPFC. Thus, we propose a trend in the hierarchical organization of the frontal cortex with respect to the planning of sequential actions: a category of sequences in the LPFC; a sequence of actions in higher motor areas; and a plan of individual actions in the primary motor and premotor cortex (Figure 3).

This concept of the hierarchical organization is consistent with Fuster’s view that the LPFC is at the top of a perception–action cycle in exerting executive functions to organize actions [3,12]. More recently, Koehlin and his colleagues [69] proposed a cascade model of cognitive control to describe the architecture of neural systems controlling actions: sensory control to select motor actions in response to sensory signals; contextual control to select stimulus–response associations; and episodic control to select task sets. How do the aforementioned properties of LPFC neurons reconcile with the cascade model? It is important to note that the episodic control is proposed as a process to select task sets in accordance with what is

Box 3. Questions for future research

Various issues must be investigated to further clarify the involvement of the LPFC in concept-based behavioral planning:

- In what way do different subregions within the LPFC take part in different aspects of executive behavioral control? The issue of a functional parcellation of the LPFC remains to be solved. On the one hand, current techniques of brain imaging in human subjects have not provided sufficient spatial resolution to reveal the exact site of measurement [70]. On the other hand, the placing of electrodes in animal studies is not generally consistent. Possible functional subdivisions along the dorsolateral axis and along the rostrocaudal axis are of much interest.
- What are the neuronal mechanisms that allow category learning with respect to action sequences? A multidisciplinary approach is required to analyze neuronal activity during learning and to develop computational models that explain the development of category specificity.
- Here, we describe category learning as an example of the abstract structure of behavior. What is the limit of such abstract representation in nonhuman primates?
- Although reward relevance is currently the most popular research topic in neuroscience, a major part of higher-order behavioral goal characterization in primate behavior is independent of the acquisition of primary reinforcers. One such example is presented here. Does this apply to other aspects of behavioral goal regulation? For instance, does the LPFC participate in developing behavioral goals that satisfy individuals at conceptual levels, leading to self-satisfaction?

dictated by events that previously occurred. In that sense, the category-specific information is interpreted as generated in the course of occurrences of events. In other words, the category information is viewed as an internally generated knowledge to exert episodic control. This view is in line with Koehlin’s model in which the LPFC, including Brodman’s area 46, takes part in episodic control of behavior.

Conclusions

Neurophysiological evidence indicates that the LPFC is primarily involved in behavioral planning that also requires cognitive information processing, but that this area is not substantially concerned with the specification of the motor aspects of behavior. In addition to behavioral factors such as task rules, task objectives, and task demands, two aspects of cognitive processing are deemed of particular interest. First, the steps required to attain a behavioral goal are processed individually in the LPFC. Second, when a large number of behavioral sequences must be planned, LPFC cells can represent categories of action sequences that possess a common temporal pattern of sequences. This representation of behavioral categories is an example of high-level processing of cognitive information that provides effective guidance for multi-phased behavioral sequences (see Box 3 for remaining, unanswered questions).

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